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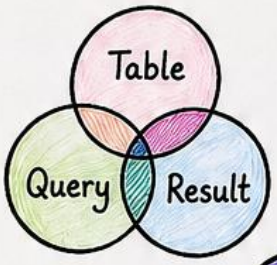
SQL

SELECT *
FROM table
WHERE condition;

HANDWRITTEN NOTES

150+ SQL QUESTIONS
FOR INTERVIEW

- [x] 150 + Interview Questions
- [x] Top Company Questions
- [x] Best Code Examples
- [x] From Basic to Advanced
- [x] Difficult Questions Included



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SQL

INTERVIEW QUESTIONS

BASIC TO ADVANCED

150+ QUESTIONS

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WHAT'S INSIDE?

01. SQL BASICS ---- 01
02. SQL QUERIES ---- 02
03. FILTERING DATA ---- 03
04. SORTING & LIMITING ---- 04
05. AGGREGATE FUNCTIONS ---- 05
06. GROUP BY & HAVING ---- 06
07. JOINS ---- 07
08. SUBQUERIES ---- 08
09. SET OPERATORS ---- 09
10. ADVANCED SQL ---- 10

COVERS

- Basic to Advanced Concepts
- Most Asked Interview Questions
- Real Interview Level Questions
- Perfect for Placement & Job Interviews
- Clean, Simple & Easy to Understand

SQL

Practice More → Code Better → Get Placed

1. SQL BASICS

SQL (Structured Query Language) is used to store, manipulate and retrieve data from relational database.

SQL is NOT case sensitive.

TYPES OF SQL COMMANDS

DDL (Data Definition Language)	DML (Data Manipulation Language)	DQL (Data Query Language)	DCL (Data Control Language)	TCL (Transaction Control Language)
CREATE, ALTER, DROP, TRUNCATE	INSERT, UPDATE, DELETE	SELECT	GRANT, REVOKE	COMMIT, ROLLBACK, SAVEPOINT

1.1 CREATE DATABASE

Creates a new database.

```
SQL> CREATE DATABASE College;
```

To use database

```
SQL> USE College;
```

1.2 CREATE TABLE

Creates a new table in the selected database.

```
SQL> CREATE TABLE Student (
        id INT PRIMARY KEY,
        name VARCHAR(50),
        age INT,
        course VARCHAR(50)
);
```

Student Table

id	name	age	course
1	Rahul	20	BCA
2	Priya	21	B.Sc
3	Amit	22	B.Tech

1.3 INSERT DATA

Inserts new records into a table.

```
SQL> INSERT INTO Student (id, name, age, course)
     VALUES (1, 'Rahul', 20, 'BCA');
```

Single quote (') is used for text values.

1.4 SELECT DATA

Retrieves data from a table.

```
SQL> SELECT * FROM Student;
```

It will return all records from Student table.



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SQL INTERVIEW QUESTIONS

BASIC TO ADVANCED

02



06 Q. What is NULL in SQL?

Ans. NULL represents missing, unknown or inapplicable data in a column.

```
-- Example
SELECT name, age
FROM Student
WHERE age IS NULL;
```

name	age
Rahul	20
Priya	NULL
Amit	22

07 Q. What is the difference between DELETE, TRUNCATE and DROP?

Ans.
- DELETE – Removes selected rows from a table.
- TRUNCATE – Removes all rows from a table, keeps structure.
- DROP – Removes table structure and all data from the table.

Command	Removes	Rollback	WHERE clause
DELETE	Selected Rows	Yes	Yes
TRUNCATE	All Rows	No	No
DROP	Table + Data	No	No



★ DELETE can be slower on large tables compared to TRUNCATE.

08 Q. What is WHERE clause?

Ans. WHERE clause is used to filter records based on a specified condition.

```
-- Example
SELECT *
FROM Student
WHERE age > 20;
```

id	name	age
1	Rahul	20
2	Priya	21
3	Amit	22

09 Q. What are the logical operators in SQL?

Ans. Logical operators are used to combine multiple conditions.

```
-- Example
SELECT * FROM Student
WHERE age > 20 AND course = 'BCA';
```

Operator	Description
AND	Returns true if both conditions are true.
OR	Returns true if at least one condition is true.
NOT	Reverses the result, returns false if the condition is true.

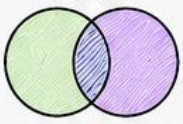


10 Q. What is the difference between UNION and UNION ALL?

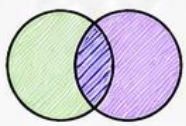
Ans.
- UNION – Combines result sets and removes duplicate rows.
- UNION ALL – Combines result sets and keeps duplicate rows.

```
-- Examples
-- UNION
SELECT name FROM Student1
UNION
SELECT name FROM Student2;
-- UNION ALL
SELECT name FROM Student1
UNION ALL
SELECT name FROM Student2;
```

UNION



UNION ALL



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03

SQL INTERVIEW QUESTIONS

BASIC TO ADVANCED



06 Q. What is the difference between DELETE, TRUNCATE and DROP?

Ans.

Feature	DELETE	TRUNCATE	DROP
Meaning	Deletes specific rows	Removes all rows from table	Deletes table completely
WHERE Clause	Yes	No	No
Rollback	Yes	No	No
Speed	Slower	Faster	Fastest
Space	Not Freed	Freed	Freed

Examples

- `DELETE FROM Student WHERE id = 1;`
- `TRUNCATE TABLE Student;`
- `DROP TABLE Student;`

07 Q. What is the use of WHERE clause?

Ans. WHERE clause is used to filter records based on specific condition.

```
SQL> SELECT * FROM Student
     WHERE age > 20;
```

id	name	age	course
2	Priya	21	B.Sc
3	Amit	22	B.Tech

08 Q. What are the comparison operators in SQL?

Ans.

=	!= or <>	>	<	>=	<=
Equal	Not Equal	Greater	Less	Greater or Equal	Less or Equal

Example

```
SELECT * FROM Student
WHERE age >= 18;
```

09 Q. How do you sort the result-set in SQL?

Ans. We use ORDER BY clause to sort the data.

- ASC → Ascending Order (Default)
- DESC → Descending Order

```
SQL> SELECT * FROM Student
     ORDER BY age DESC;
```



10 Q. What is LIMIT clause?

Ans. LIMIT is used to restrict the number of records returned.

```
SQL> SELECT * FROM Student
     LIMIT 3;
```

id	name	age	course
1	Rahul	20	BCA
2	Priya	21	B.Sc
3	Amit	22	B.Tech

LIMIT is very useful in interviews for large tables.



SQL INTERVIEW QUESTIONS

BASIC TO ADVANCED



11 Q. What are the comparison operators in SQL?

Ans Comparison operators are used to compare two values.

```
-- Example
SELECT name, age
FROM Student
WHERE age >= 20;
```

Operator	Meaning	Example
=	Equal to	age = 20
<> or !=	Not equal to	age <> 20
>	Greater than	age > 20
<	Less than	age < 20
>=	Greater than or equal to	age >= 20
<=	Less than or equal to	age <= 20



12 Q. What is the difference between ORDER BY and GROUP BY?

Ans

- ORDER BY – Used to sort the result set in ascending (ASC) or descending (DESC) order.
- GROUP BY – Used to group rows that have the same values into summary rows, often used with aggregate functions.

```
-- ORDER BY Example
SELECT name, age
FROM Student
ORDER BY age DESC;
```

```
-- GROUP BY Example
SELECT course, COUNT(*)
FROM Student
GROUP BY course;
```

Before ORDER BY (age)

name	age
Amit	22
Rahul	20
Priya	21

⇨

After ORDER BY age DESC

name	age
Amit	22
Priya	21
Rahul	20

GROUP BY course

course	Total_Students
BCA	2
B.Sc	1
B.Tech	1

13 Q. What are aggregate functions in SQL? Give examples.

Ans Aggregate functions perform a calculation on a set of values and return a single value.

```
-- Example
SELECT COUNT(*), AVG(age),
       MAX(age), MIN(age)
FROM Student;
```

Function	Description	Example
COUNT()	Counts the number of rows	COUNT(*)
SUM()	Returns the total sum	SUM(salary)
AVG()	Returns the average value	AVG(age)
MIN()	Returns the minimum value	MIN(age)
MAX()	Returns the maximum value	MAX(salary)



14 Q. What is the difference between INNER JOIN and LEFT JOIN?

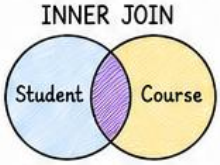
Ans

- INNER JOIN – Returns only the matching rows from both tables.
- LEFT JOIN – Returns all rows from the left table and matched rows from the right table. If no match, NULL is returned.

```
-- INNER JOIN Example
SELECT s.name, c.course
FROM Student s
INNER JOIN Course c
ON s.course_id = c.id;
```

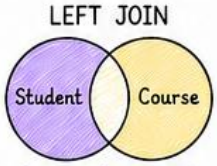
```
-- LEFT JOIN Example
SELECT s.name, c.course
FROM Student s
LEFT JOIN Course c
ON s.course_id = c.id;
```

INNER JOIN



Only Matching Rows

LEFT JOIN



All from Left + Matching



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SQL INTERVIEW QUESTIONS

BASIC TO ADVANCED

04



---

20 Q. What is the difference between CROSS JOIN and INNER JOIN?

Ans.
- CROSS JOIN – Returns the Cartesian product of both tables.
- INNER JOIN – Returns only matching rows based on the join condition.

```
-- CROSS JOIN Example
SELECT *
FROM A
CROSS JOIN B;
```

Table A

id
1
2

×

Table B

name
X
Y

⇨ Result (4 rows)

id	name
1	X
1	Y
2	X
2	Y

```
-- INNER JOIN Example
SELECT *
FROM A
INNER JOIN B ON A.id = B.id;
```

Table A

id
1
2
3

⋈

Table B

id
2
3
4

⇨ Result (2 rows)

id
2
3

---

21 Q. What is the difference between DELETE, TRUNCATE and DROP?

Ans.

Feature	DELETE	TRUNCATE	DROP
Type	DML	DDL	DDL
Removes	Rows	All Rows	Table + Data
WHERE Clause	Yes	No	No
Rollback	Yes	No	No
Speed	Slower	Faster	Fastest
Identity/Auto Increment	Not Resets	Resets	Not Applicable
Space	Space not freed	Space freed	Space freed
Structure	Same	Same	Removed

```
-- Examples
```
- DELETE FROM Student WHERE age < 18;
- TRUNCATE TABLE Student;
- DROP TABLE Student;



Note:
- DELETE can be rolled back.
- TRUNCATE and DROP cannot be rolled back in most databases.

---

22 Q. What is a subquery? Give types of subqueries with examples.

Ans. A subquery is a query inside another query.

Types of Subqueries:

1 Single Row Subquery

Returns a single value.

```
-- Example
SELECT name
FROM Student
WHERE age = (SELECT MAX(age)
             FROM Student);
```

2 Multiple Row Subquery

Returns multiple rows.

```
-- Example
SELECT name
FROM Student
WHERE age IN (SELECT age
              FROM Student
              WHERE course = 'BCA');
```

3 Multiple Column Subquery

Returns multiple columns.

```
-- Example
SELECT name, course
FROM Student
WHERE (course, age) IN
 (SELECT course, MAX(age)
  FROM Student
  GROUP BY course);
```

4 Correlated Subquery

A subquery that depends on the outer query.

```
-- Example
SELECT name
FROM Student s1
WHERE age > (SELECT AVG(age)
   FROM Student s2
   WHERE s1.course = s2.course);
```

---

23 Q. What is the difference between DISTINCT and GROUP BY?

Ans.
- DISTINCT – Used to return unique (distinct) values from a column.
- GROUP BY – Used to group rows that have the same values into summary rows, often used with aggregate functions.

Pro Tip:
Use DISTINCT when you just need unique values.
Use GROUP BY when you need calculations.



```
-- DISTINCT Example
SELECT DISTINCT course
FROM Student;
```

Result

course
BCA
B.Sc
B.Tech

```
-- GROUP BY Example
SELECT course, COUNT(*)
FROM Student
GROUP BY course;
```

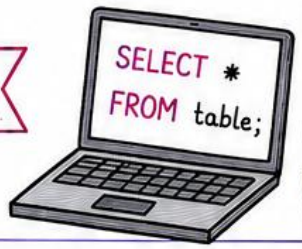
Result

course	COUNT(*)
BCA	25
B.Sc	15
B.Tech	10

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SQL INTERVIEW QUESTIONS



BASIC LEVEL QUESTIONS

25 Q. What is a table in SQL?

Ans. A table is a collection of related data organized in rows and columns.

Example:

Student Table

id	name	age	course
1	Rahul	20	BCA
2	Priya	21	B.Sc
3	Amit	22	B.Tech

---

26 Q. What is a primary key?

Ans.
- A primary key uniquely identifies each row in a table.
- It cannot contain NULL values.
- Only one primary key can exist in a table.

Example:

```
CREATE TABLE Student (
  id INT PRIMARY KEY,
  name VARCHAR(50),
  age INT
);
```

Key Points
- ☆ Unique
- Not NULL
- One per table

---

27 Q. What is a foreign key?

Ans.
- A foreign key is a column (or group of columns) in one table that refers to the primary key in another table.
- It helps to maintain referential integrity.

Example:

```
CREATE TABLE Orders (
  order_id INT PRIMARY KEY,
  student_id INT,
  FOREIGN KEY (student_id)
  REFERENCES Student(id)
);
```

Student Table
- id (PK)
- name
- age

Orders Table
- order_id (PK)
- student_id (FK)

---

28 Q. What is the difference between CHAR and VARCHAR?

Ans.

Feature	CHAR	VARCHAR
Length	Fixed length	Variable length
Space	Uses fixed space	Uses required space
Performance	Faster	Slower (slightly)
Example	CHAR(10)	VARCHAR(10)

Example:

```
CREATE TABLE Test (
  name CHAR(10),    -- Fixed
  city VARCHAR(10)  -- Variable
);
```

Note
Use CHAR when size is fixed. Use VARCHAR when size varies.

---

29 Q. What is the difference between DELETE, TRUNCATE and DROP?

Ans.

Feature	DELETE	TRUNCATE	DROP
Type	DML	DDL	DDL
Removes	Rows	All rows	Table + Data
WHERE Clause	Yes	No	No
Rollback	Yes	No	No
Auto Increment	Not reset	Reset	Not applicable
Structure	Same	Same	Removed

Example:
- `DELETE FROM Student WHERE age < 18;`
- `TRUNCATE TABLE Student;`
- `DROP TABLE Student;`

☆ Note
- DELETE is slower on large tables.
- TRUNCATE is faster than DELETE.

---

30 Q. What is the difference between WHERE and HAVING?

Ans.

Feature	WHERE	HAVING
Used with	WHERE clause	HAVING clause
Works on	Individual rows	Groups
Used with GROUP BY	No	Yes
Example	`SELECT * FROM Student WHERE age > 20;`	`SELECT course, COUNT(*) FROM Student GROUP BY course HAVING COUNT(*) > 10;`

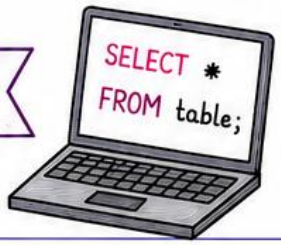
Remember
- WHERE filters rows before grouping.
- HAVING filters groups after grouping.

Quick Tip
WHERE → Before GROUP BY
HAVING → After GROUP BY



SQL INTERVIEW QUESTIONS

BASIC LEVEL QUESTIONS



31 Q. What is the difference between UNION and UNION ALL?

Ans.

- **UNION** – Combines result sets and removes duplicate rows.
- **UNION ALL** – Combines result sets and keeps all duplicate rows.

Example:

```
-- UNION Example
SELECT name FROM Student1
UNION
SELECT name FROM Student2;
```

Result (Duplicates Removed)

name
Amit
Rahul
Priya

```
-- UNION ALL Example
SELECT name FROM Student1
UNION ALL
SELECT name FROM Student2;
```

Result (Duplicates Kept)

name
Amit
Rahul
Amit
Priya

Key Point

Use UNION when you want unique values. Use UNION ALL when you want all values (including duplicates).

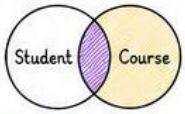
32 Q. What is the difference between INNER JOIN and LEFT JOIN?

Ans.

- **INNER JOIN** – Returns only matching rows from both tables.
- **LEFT JOIN** – Returns all rows from left table and matching rows from right table. If no match, NULL is returned.

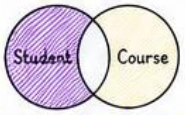
Example:

```
-- INNER JOIN Example
SELECT s.name, c.course
FROM Student s
INNER JOIN Course c
ON s.course_id = c.id;
```



Only Matching Rows

```
-- LEFT JOIN Example
SELECT s.name, c.course
FROM Student s
LEFT JOIN Course c
ON s.course_id = c.id;
```



All from Left + Matching

Note

If no match in right table (Course), columns from right table will be NULL.

33 Q. What is the difference between GROUP BY and ORDER BY?

Ans.

- **GROUP BY** – Groups rows that have same values into summary rows.
- **ORDER BY** – Sorts the result set in ascending (ASC) or descending (DESC) order.

Example:

```
-- GROUP BY Example
SELECT course, COUNT(*)
FROM Student
GROUP BY course;
```

Result

course	COUNT(*)
BCA	3
B.Sc	2
B.Tech	1

```
-- ORDER BY Example
SELECT name, age
FROM Student
ORDER BY age DESC;
```

Result

name	age
Amit	22
Priya	21
Rahul	20

Remember

GROUP BY is used with aggregate functions like COUNT(), SUM(), AVG(), etc.

ORDER BY is used to sort the final result.

34 Q. What is the difference between DISTINCT and ALL?

Ans.

- **DISTINCT** – Returns only unique values.
- **ALL** – Returns all values including duplicates. (ALL is default in SQL)

Example:

```
-- DISTINCT Example
SELECT DISTINCT city
FROM Student;
```

Result

city
Delhi
Mumbai
Pune

```
-- ALL Example
SELECT city
FROM Student;
```

Result

city
Delhi
Mumbai
Delhi
Pune

Key Point

Use DISTINCT when you want unique values only.

35 Q. What are aggregate functions in SQL? Give examples.

Ans. Aggregate functions perform a calculation on a set of values and return a single value.

Common Aggregate Functions

- COUNT() – Counts number of rows
- SUM() – Returns total sum
- AVG() – Returns average
- MIN() – Returns minimum value
- MAX() – Returns maximum value

Example:

```
SELECT COUNT(*), SUM(age), AVG(age),
  MIN(age), MAX(age)
FROM Student;
```

Result

COUNT(*)	SUM(age)	AVG(age)	MIN(age)	MAX(age)
6	123	20.50	18	22

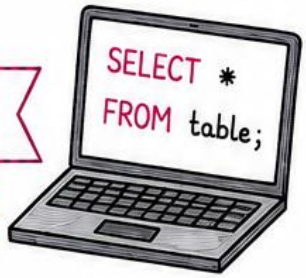
Tip

COUNT(*) counts all rows including NULLs. COUNT(column) counts only non-NULL values.



SQL INTERVIEW QUESTIONS

INTERMEDIATE LEVEL QUESTIONS





36 Q. How do you find the 2nd highest salary from the Employee table?

Ans. Use ORDER BY to sort in descending order and LIMIT to get the 2nd row.

Example:

```
SELECT salary
FROM Employee
ORDER BY salary DESC
LIMIT 1 OFFSET 1;
```

Note

- ★ OFFSET 1 skips the highest salary.
- LIMIT 1 returns the next row (2nd highest).

---

37 Q. Write a query to find duplicate emails from the Users table.

Ans. Group by email and use HAVING to filter those having count > 1.

Example:

```
SELECT email, COUNT(*) AS cnt
FROM Users
GROUP BY email
HAVING COUNT(*) > 1;
```

Result:

email	cnt
a@gmail.com	3
b@gmail.com	2
c@gmail.com	2

---

38 Q. How do you fetch the top 3 records from a table?

Ans. Use LIMIT clause to restrict the number of rows returned.

Example:

```
SELECT *
FROM Products
ORDER BY price DESC
LIMIT 3;
```

Result (Top 3 Expensive Products):

id	name	price
101	Laptop	80000
102	Smartphone	60000
103	Tablet	45000

---

39 Q. How do you find employees who earn more than the average salary?

Ans. Use subquery to calculate average salary first, then compare.

Example:

```
SELECT name, salary
FROM Employee
WHERE salary > (SELECT AVG(salary)
      FROM Employee);
```

Note

- ★ Subquery calculates average salary from the same table.

---

40 Q. Write a query to find employees who joined in the same year.

Ans. Extract year from the date and group by it.

Example:

```
SELECT YEAR(joining_date) AS join_year,
    COUNT(*) AS total_employees
FROM Employee
GROUP BY YEAR(joining_date);
```

Result:

join_year	total_employees
2022	15
2023	22
2024	18

Tip

YEAR(), MONTH(), DAY() functions help in extracting date parts.



SQL INTERVIEW QUESTIONS

ADVANCED LEVEL QUESTIONS



41 Q. Write a query to find the 2nd highest salary in the company.

Ans.
- Use ORDER BY salary DESC and LIMIT 1 OFFSET 1.
- This skips the highest salary and gives the 2nd highest.

Example:

```
SELECT salary
FROM employee
ORDER BY salary DESC
LIMIT 1 OFFSET 1;
```

Result:

salary
50000

Note: You can also use DENSE_RANK() to get the 2nd highest salary.

---

42 Q. How do you find the nth highest salary using RANK() or DENSE_RANK()?

Ans.
- Use RANK() or DENSE_RANK() with ORDER BY salary DESC.
- Then filter where rank = n.

Example (3rd highest salary):

```
SELECT salary
FROM (
  SELECT salary,
    RANK() OVER(ORDER BY salary DESC) AS rnk
  FROM employee
) t
WHERE rnk = 3;
```

Result:

salary
40000

RANK() gives same rank for ties, DENSE_RANK() removes gaps.

---

43 Q. Write a query to get the count of employees in each department.

Ans.
- Use GROUP BY department and COUNT().
- This will give total employees per department.

Example:

```
SELECT department, COUNT(*) AS emp_count
FROM employee
GROUP BY department;
```

Result:

department	emp_count
HR	5
IT	8
Sales	6
Marketing	4

---

44 Q. How do you find duplicate records in a table? (e.g., duplicate emails)?

Ans.
- Use GROUP BY and HAVING COUNT(*) > 1.
- This will give the values that appear more than once.

Example:

```
SELECT email, COUNT(*) AS cnt
FROM users
GROUP BY email
HAVING COUNT(*) > 1;
```

Result:

email	cnt
a@gmail.com	3
b@gmail.com	2
c@gmail.com	2

Note: You can also use a subquery or window function (ROW_NUMBER()) to find duplicates.

---

45 Q. Write a query to find employees who have worked in the same department for more than 5 years.

Ans.
- Use a self-join or subquery with date difference.
- Compare joining_date + 5 years with current date.

Example (using subquery):

```
SELECT e1.name, e1.department
FROM employee e1
WHERE e1.joining_date <= DATE_SUB(CURDATE(),
                                  INTERVAL 5 YEAR)
  AND e1.department IN (
      SELECT department
      FROM employee
      WHERE DATEDIFF(CURDATE(), joining_date) > 5
);
```

Note: You can also use DATEDIFF() or TIMESTAMPDIFF() for date comparison (according to your DB).



SQL INTERVIEW QUESTIONS

INTERMEDIATE LEVEL QUESTIONS



46

Q. Find the longest name in the Employee table.

Ans. Use ORDER BY with LENGTH() in DESC order and LIMIT 1.

Example :

```
SELECT name
FROM Employee
ORDER BY LENGTH(name) DESC
LIMIT 1;
```

Result :

name
Shubham Verma

Note

★ LENGTH() returns the number of characters in a string.

47

Q. Write a query to find the second lowest salary from the Employee table.

Ans. Use DISTINCT to remove duplicates and LIMIT with OFFSET.

Example :

```
SELECT DISTINCT salary
FROM Employee
ORDER BY salary ASC
LIMIT 1 OFFSET 1;
```

Result :

salary
35000

Note

★ OFFSET 1 skips the lowest salary and returns the next (second lowest).

48

Q. Write a query to find employees who earn more than their department average salary.

Ans. Use subquery to compare salary with department average.

Example :

```
SELECT e.name, e.department, e.salary
FROM Employee e
WHERE e.salary > (
    SELECT AVG(salary)
    FROM Employee e2
    WHERE e2.department = e.department
);
```

Result (Sample) :

name	department	salary
Amit	IT	60000
Priya	HR	55000
Karan	IT	65000

Note

★ Compares each employee's salary with average salary of the same department.

49

Q. Write a query to find departments where the total salary expense is greater than 2,00,000.

Ans. Use GROUP BY to calculate total salary and HAVING to filter.

Example :

```
SELECT department, SUM(salary) AS total_salary
FROM Employee
GROUP BY department
HAVING SUM(salary) > 200000;
```

Result :

department	total_salary
IT	315000
Sales	245000

Tip

HAVING is used to filter groups. WHERE cannot be used with aggregate functions.

50

Q. Write a query to find the month in which the company spent the most on salaries.

Ans. Extract month from date, group by month and find max total salary.

Example :

```
SELECT MONTH(joining_date) AS month,
       SUM(salary) AS total_salary
FROM Employee
GROUP BY MONTH(joining_date)
ORDER BY total_salary DESC
LIMIT 1;
```

Result :

month	total_salary
7	285000

★ MONTH() extracts month number from date (1-12).

★ Key Takeaway

Practice different SQL patterns like subqueries, aggregation, grouping, and filtering.

Keep Practicing

- Read the question carefully
- Break the problem logically
- Write, test & optimize your queries

Practice More ★ Code Better
Get Placed
@ code.withnarayan